From Brownian motion to Bessel processes: Proposition 1.21 in Lawler

Debeilei Wang

1 A theorem we know

In Analysis III we proved the following.

Theorem 1 (Theorem 161)**.** *d-dimensional Brownian motion is point recurrent for $d = 1$, neighborhood recurrent (but not point recurrent) for $d = 2$, and transient for $d \geq 3$.*

Proposition 1.21 in Lawler's *Conformally Invariant Processes in the Plane* is a statement of the same flavor for *Bessel processes*, defined by an SDE (see below); for integer d a Bessel process has the law of the radial part of d-dimensional Brownian motion, but the dimension d is allowed to be any real number. The plan of this note:

1. explain how Theorem 1 translates to the radial part, and **conjecture** what should be true for a Bessel process (§2–3);
2. **verify** the conjecture by proving the first four parts of Proposition 1.21, organized by the three quantities T_x, $\inf_t X_t$, $\lim_t X_t$ (§4–8).

2 Bessel processes as radial parts of Brownian motion

Let W be a d-dimensional standard Brownian motion started at a point of norm $x > 0$, and let $X_t = \|W_t\|$. By Itô's formula, up to the time T_x at which X hits 0,

$$dX_t = \frac{d-1}{2X_t}\,dt + d\beta_t, \qquad \beta_t = \int_0^t \frac{W_s}{\|W_s\|} \cdot dW_s,$$

and β is a one-dimensional Brownian motion (Lévy's characterization). So X solves

$$dX_t = \frac{a}{X_t}\,dt + d\beta_t, \qquad a = \frac{d-1}{2},$$

the **Bessel process** with parameter a. This SDE makes sense for every real $d > 0$ (we assume $a > -\frac{1}{2}$ below), not only for integers. **In the proofs below, a Bessel process is by definition a solution of this SDE; the norm interpretation is never used.** For integer d the norm $\|W\|$ has the same *law* as the solution; note also that the coupling used below (one Brownian motion B driving X^x for all x) is a coupling of SDE solutions, not the norms of translates of a single Brownian path. Note that $a < \frac{1}{2} \iff d < 2$, $a = \frac{1}{2} \iff d = 2$, $a > \frac{1}{2} \iff d > 2$.

Dictionary (integer d). Each statement below is an almost sure property of the path of $\|W\|$, hence depends only on its law; so a.s. properties of a Bessel process with a fixed starting point transfer to $\|W\|$.

Brownian motion W	Radial part $X = \|W\|$ (same law as the Bessel process)
W hits the origin	X hits 0: $T_x < \infty$
W returns to every neighborhood of the origin	$\inf_t X_t = 0$ (and $\liminf_{t\to\infty} X_t = 0$)
$\|W_t\| \to \infty$	$X_t \to \infty$

Point recurrence for $d = 1$ implies that W hits the origin. Point *transience* for $d = 2$ (a given point is almost surely not hit) gives $T_x = \infty$. Neighborhood recurrence gives $\liminf X_t = 0$. For $d \geq 3$, transience is $X_t \to \infty$. (Statements about hitting a point *other* than the origin, or about the angle, have no counterpart for X.)

3 Conjecture and the statement

Reading Theorem 1 through the dictionary, with $d = 2a + 1$, we are led to guess the following behavior of a Bessel process X^x started at $x > 0$:

- $a < \frac{1}{2}$ ($d < 2$, like $d = 1$): X hits the origin in finite time.
- $a = \frac{1}{2}$ ($d = 2$): X never hits the origin but comes arbitrarily close to it infinitely often: neighborhood recurrence at the origin. Whether X *also* goes to high levels infinitely often, or could stay in a bounded region oscillating between 0 and an upper barrier, is not clear from this reading: we mark it by **?** and decide it in the proof.
- $a > \frac{1}{2}$ ($d > 2$): X is transient: it never hits the origin, stays away from it, and escapes to infinity.

regime	T_x	$\inf_t X_t$	$\lim_t X_t$	$\liminf$ / $\limsup$
$a < \frac{1}{2}$	$< \infty$	0 (attained at T_x)	0 as $t \uparrow T_x$	0
$a = \frac{1}{2}$	∞	0	**?**	0 / **?**
$a > \frac{1}{2}$	∞	> 0	∞	∞

Table 1. Conjecture. The entries marked **?** are undecided at this point.

The last column is redundant when the limit exists. For $a < \frac{1}{2}$ the process is only defined up to T_x and all statements are as $t \uparrow T_x$.

Theorem 2 (Lawler, Proposition 1.21 (first four parts))**.** *Let $a > -\frac{1}{2}$. With probability one, simultaneously for all $x > 0$:*

(1) if $a \geq \frac{1}{2}$, then $T_x = \infty$;
(2) if $a = \frac{1}{2}$, then $\inf_t X^x_t = 0$;
(3) if $a > \frac{1}{2}$, then $X^x_t \to \infty$;
(4) if $a < \frac{1}{2}$, then $T_x < \infty$.

The three parts of the proof follow the three columns of Table 1: **Part I** is about T_x, **Part II** about $\inf_t X_t$, **Part III** about $\lim_t X_t$; in Part III we also settle the undecided entries of Table 1.

4 Setting, notation, comparison

Let B be a standard Brownian motion, $B_0 = 0$. For $x > 0$,

$$dX_t^x = \frac{a}{X_t^x}\,dt + dB_t, \quad X_0^x = x, \qquad X_t^x = x + B_t + a\int_0^t \frac{ds}{X_s^x} \quad (t < T_x),$$

where $T_x := \inf\{t : X_t^x = 0\}$ is the zero time of X^x. **All X^x are driven by the same B.** (Path by path, $Y = X^x - B$ solves the deterministic equation $Y' = a/(Y + B_t(\omega))$, so all the X^x can be built simultaneously for every ω.) $\mathbf{P}^z, \mathbf{E}^z$ refer to the process started at z.

For a fixed path $X = X^x$ and a level $y > 0$ let

$$\tau_y = \inf\{t : X_t = y\} \quad (\inf\emptyset = \infty), \qquad \tau_0 := T_x.$$

Convention: T with a starting point as subscript is a zero time; τ with a level as subscript is a hitting time of a level by one fixed path. For $0 \le y < x < R$:

(P1) $\tau_y \le \tau_0$; $y \mapsto \tau_y$ is non-increasing on $[0, x)$; $R \mapsto \tau_R$ is non-decreasing.

(P2) If $\tau_y < \infty$, then X is bounded on $[0, \tau_y]$, so $\tau_R = \infty > \tau_y$ for large R. Hence $\{\tau_y < \infty\} = \bigcup_n \{\tau_y < \tau_n\}$, an increasing union.

Theorem 3 (Comparison)**.** *If $0 < x < y$, then for* every *ω: $X_t^x < X_t^y$ for $t < T_x$, and $T_x \le T_y$.*

Proof. For $t < T_x$ put $D_t = X_t^y - X_t^x$. The Brownian parts cancel, so

$$D_t' = a\Big(\frac{1}{X_t^y} - \frac{1}{X_t^x}\Big) = -\frac{a\,D_t}{X_t^x X_t^y}, \qquad D_0 = y - x > 0,$$

hence $D_t = (y-x)\exp\big(-a\int_0^t \frac{ds}{X_s^x X_s^y}\big) > 0$. If $T_y < T_x$, then $D_{T_y} = -X_{T_y}^x < 0$, while $D \ge 0$ on $[0, T_y]$ by continuity. So $T_x \le T_y$. □

Exit probabilities

Lemma 1 (Exit probabilities)**.** *Let $0 < y_1 < z < y_2 < \infty$ and $\sigma = \tau_{y_1} \wedge \tau_{y_2}$ for the process started at z. Then $\mathbf{E}^z[\sigma] < \infty$ and*

$$\mathbf{P}^z\{\tau_{y_2} < \tau_{y_1}\} = \phi_0(z) := \begin{cases} \dfrac{z^{1-2a} - y_1^{1-2a}}{y_2^{1-2a} - y_1^{1-2a}}, & a \ne \frac12, \\[2ex] \dfrac{\log z - \log y_1}{\log y_2 - \log y_1}, & a = \frac12. \end{cases}$$

Proof. By Itô, $d(X_t^2) = (2a+1)\,dt + 2X_t\,dB_t$. Up to σ the process lies in $[y_1, y_2]$, so $(2a+1)\mathbf{E}[t\wedge\sigma] = \mathbf{E}[X_{t\wedge\sigma}^2] - z^2 \le y_2^2$; hence $\mathbf{E}\sigma \le y_2^2/(2a+1) < \infty$ and $\sigma < \infty$ a.s. Next $\phi_0 \in C^2$, $\phi_0(y_1) = 0$, $\phi_0(y_2) = 1$, $\phi_0'(u) = cu^{-2a}$ with $c > 0$, so $\frac12\phi_0'' + \frac{a}{u}\phi_0' = 0$. By Itô, $M_t = \phi_0(X_{t\wedge\sigma})$ is a bounded martingale, and optional sampling gives $\phi_0(z) = \mathbf{E}M_\sigma = \mathbf{P}^z\{X_\sigma = y_2\}$. □

Lemma 2 (The limits we need)**.** *Let $0 < y < z < R$ and $p(y,R) = \mathbf{P}^z\{\tau_y < \tau_R\}$. By Lemma 1, $p(y,R) = \frac{R^{1-2a} - z^{1-2a}}{R^{1-2a} - y^{1-2a}}$ for $a \ne \frac12$ and $p(y,R) = \frac{\log R - \log z}{\log R - \log y}$ for $a = \frac12$. Hence*

	$a < \frac12$	$a = \frac12$	$a > \frac12$
$\lim_{y\to 0^+} p(y,R)$	$1 - (z/R)^{1-2a}$	0	0
$\lim_{R\to\infty} p(y,R)$	1	1	$(y/z)^{2a-1}$

Consequently:
(a) $\mathbf{P}^z\{\tau_y < \infty\} = \lim_{R\to\infty} p(y,R)$ *is* 1 *if* $a \le \frac{1}{2}$ *and* $(y/z)^{2a-1}$ *if* $a > \frac{1}{2}$ *(by (P2));*
(b) if $a \ge \frac{1}{2}$*, then* $\mathbf{P}^z\{\tau_R < \infty\} = 1$ *for all* $R > z$*.*

Proof. The table is the computation of limits ($1 - 2a < 0$ for $a > \frac{1}{2}$, so $y^{1-2a} \to \infty$ as $y \to 0$ and $R^{1-2a} \to 0$ as $R \to \infty$). (a) is the second row with (P2). (b): $\{\tau_R < \infty\} \supseteq \{\tau_R < \tau_y\}$ for every $y < z$, and $\mathbf{P}^z\{\tau_R < \tau_y\} = 1 - p(y,R) \to 1$ as $y \to 0$ when $a \ge \frac{1}{2}$. □

From a fixed starting point to all starting points

Lemma 3 (Rational starting points)**.** *Suppose that, for each fixed* $x > 0$*, one of the conclusions (1)–(4) holds a.s. Then it holds a.s. simultaneously for all* $x > 0$*.*

Proof. Intersect over rational starting points to get one event Ω_0 of probability one on which the conclusion holds for all rational q. Fix $\omega \in \Omega_0$ and a real $x > 0$.

- (1) $T_x = \infty$: take rational $q < x$; by Theorem 3, $T_x \ge T_q = \infty$.
- (2) $\inf X^x = 0$: by (1), $T_x = \infty$; take rational $q > x$; then $0 < X_t^x < X_t^q$, so $\inf X^x \le \inf X^q = 0$.
- (3) $X_t^x \to \infty$: take rational $q < x$; $X_t^x > X_t^q \to \infty$.
- (4) $T_x < \infty$: take rational $q > x$; $T_x \le T_q < \infty$.

□

By Lemma 3 we fix $x > 0$ and write $X = X^x$ from now on.

Part I. The zero time T_x

Notation. For $R > x$ let

$$E_R := \bigcap_{0<y<x} \{\tau_y < \tau_R\} = \bigcap_{n>1/x} \{\tau_{1/n} < \tau_R\},$$

"the path reaches every level $y \in (0,x)$ before the level R" (the two intersections agree because $y \mapsto \tau_y$ is non-increasing, (P1)). Two facts:

- E_R increases with R (since $R \mapsto \tau_R$ is non-decreasing);
- $\mathbf{P}(E_R) = \lim_{y\to 0} p(y,R)$, since the events $\{\tau_y < \tau_R\}$ decrease as $y \downarrow 0$.

Hence $\mathbf{P}\big(\bigcup_{R>x} E_R\big) = \lim_{R\to\infty}\lim_{y\to 0} p(y,R) =: \Lambda(a)$, and by the table in Lemma 2,

$$\Lambda(a) = 0 \ \ (a \ge \tfrac{1}{2}), \qquad \Lambda(a) = \lim_{R\to\infty}\Big[1 - \Big(\frac{x}{R}\Big)^{1-2a}\Big] = 1 \ \ (a < \tfrac{1}{2}).$$

Step 1: a union–intersection and an upper bound

Idea. To reach 0 the path must first pass every level $y < x$, so $\{T_x < \infty\}$ is contained in the union–intersection $\bigcup_R E_R$.

Lemma 4. *For every* a*,* $\{T_x < \infty\} \subseteq \bigcup_{R>x} E_R$ *(pathwise). Hence* $\mathbf{P}\{T_x < \infty\} \le \Lambda(a)$*.*

Proof. By (P2), $\{\tau_0 < \infty\} = \bigcup_n \{\tau_0 < \tau_n\}$ (increasing). For every $0 < y < x$, $\tau_y \le \tau_0$ by (P1), so $\{\tau_0 < \tau_R\} \subseteq \{\tau_y < \tau_R\}$ for all such y, i.e. $\{\tau_0 < \tau_R\} \subseteq E_R$. Taking the union over R gives the inclusion, and $\mathbf{P}\{T_x < \infty\} \le \mathbf{P}(\bigcup_R E_R) = \Lambda(a)$. □

(1) $a \geq \frac{1}{2} \;\Rightarrow\; T_x = \infty$ **a.s.**

Here $\Lambda(a) = 0$, so Lemma 4 gives $\mathbf{P}\{T_x < \infty\} = 0$. The upper bound is all we need.

(4) $a < \frac{1}{2} \;\Rightarrow\; T_x < \infty$ **a.s.**

For $a < \frac{1}{2}$ we have $\Lambda(a) = 1$: the upper bound of Lemma 4 is the trivial bound $\mathbf{P}\{T_x < \infty\} \leq 1$ and tells us nothing. We need a *lower* bound, that is, the reverse inclusion $\bigcup_R E_R \subseteq \{T_x < \infty\}$ up to a null set.

Remark (The reverse inclusion is not obvious). It fails path by path: a path may satisfy $\tau_{1/n} < \tau_R$ for all n and still never reach 0, if the times $\tau_{1/n}$ increase to ∞. Set $\tau_\infty := \lim_{n\to\infty} \tau_{1/n}$ (non-decreasing). The point is to show that this bad event has probability 0.

Lemma 5 (Claim)**.** *Let $a < \frac{1}{2}$ and $R > x$. Then $\mathbf{P}\big(E_R \cap \{\tau_\infty = \infty\}\big) = 0$.*

Proof. Case $a \leq 0$. For $t < T_x$, $X_t = x + B_t + a\int_0^t ds/X_s \leq x + B_t$. On $\{T_x = \infty\}$ this gives $0 < X_t \leq x + B_t$ for all t, hence $B_t > -x$ for all t, an event of probability 0 because $\liminf_{t\to\infty} B_t = -\infty$ a.s. So $T_x < \infty$ a.s., and since $\tau_{1/n} \leq \tau_0$, $\tau_\infty \leq \tau_0 < \infty$ a.s.; the claim is trivial.

Case $0 < a < \frac{1}{2}$. First, if $\tau_R < \infty$, then on E_R we have $\tau_{1/n} < \tau_R$ for all n, so $\tau_\infty \leq \tau_R < \infty$; hence $E_R \cap \{\tau_R < \infty\} \cap \{\tau_\infty = \infty\} = \emptyset$. Next, suppose $\tau_R = \infty$ and $\tau_\infty = \infty$. For every t there is n with $\tau_{1/n} > t$, so $X_s > 1/n > 0$ on $[0, t]$. Hence $T_x = \infty$ and $X_t = x + B_t + a\int_0^t ds/X_s \geq x + B_t$ for all t (as $a > 0$). Since $\tau_R = \infty$ and $X_0 = x < R$, also $X_t < R$ for all t. So $B_t < R - x$ for all $t \geq 0$. Thus

$$E_R \cap \{\tau_R = \infty\} \cap \{\tau_\infty = \infty\} \subseteq \{B_t < R - x \;\; \forall t \geq 0\},$$

and the right side has probability 0 because $\limsup_{t\to\infty} B_t = \infty$ a.s. □

Lemma 6. *Let $a < \frac{1}{2}$ and $R > x$. Then $E_R = \{T_x < \tau_R\}$ up to a null set.*

Proof. "$\supseteq$" holds path by path ((P1): $\{\tau_0 < \tau_R\} \subseteq E_R$). For "$\subseteq$", consider $\omega \in E_R \cap \{\tau_\infty < \infty\}$; by Lemma 5 this set differs from E_R by a null set.

- $\tau_\infty = \tau_0$: $\tau_{1/n} \leq \tau_0$ gives $\tau_\infty \leq \tau_0$; if $\tau_\infty < \tau_0$, continuity at τ_∞ would give $X_{\tau_\infty} = \lim_n X_{\tau_{1/n}} = \lim_n 1/n = 0$, contradicting $X > 0$ on $[0, \tau_0)$.
- $\tau_\infty < \tau_R$: we have $\tau_\infty \leq \tau_R$ (as $\tau_{1/n} < \tau_R$); if $\tau_R < \infty$ and $\tau_\infty = \tau_R$, then $X_{\tau_\infty} = 0 \neq R = X_{\tau_R}$; if $\tau_R = \infty$, then $\tau_\infty < \infty = \tau_R$.

So $\tau_0 = \tau_\infty < \tau_R$, i.e. $\omega \in \{T_x < \tau_R\}$. □

Proof of (4). By (P2) and Lemma 6, a.s. $\{T_x < \infty\} = \bigcup_R \{T_x < \tau_R\} = \bigcup_R E_R$, so

$$\mathbf{P}\{T_x < \infty\} = \mathbf{P}\Big(\bigcup_R E_R\Big) = \Lambda(a) = 1. \qquad \square$$

The result for every a: an equality, not just an inclusion

Theorem 4. *For every $a > -\frac{1}{2}$, almost surely*

$$\{T_x < \infty\} \;=\; \bigcup_{R>x} \bigcap_{0<y<x} \{\tau_y < \tau_R\}, \qquad \textit{hence} \qquad \mathbf{P}\{T_x < \infty\} = \lim_{R\to\infty} \lim_{y\to 0^+} \mathbf{P}^x\{\tau_y < \tau_R\}.$$

Proof. For $a \geq \frac{1}{2}$ both sides are null sets (Lemma 4 and $\Lambda = 0$). For $a < \frac{1}{2}$ this is the proof of (4). The second statement follows from $\mathbf{P}(E_R) = \lim_{y\to 0} p(y, R)$ and the monotonicity of $R \mapsto E_R$. □

Reading off (1) and (4). The path-level identity of Theorem 4 is the same for all a; the two assertions differ only in the value of the iterated limit of the explicit exit probabilities (Lemma 2):

	$\lim_{y\to 0} p(y,R)$	$\mathbf{P}\{T_x<\infty\} = \lim_R \lim_y p(y,R)$
$a \geq \frac{1}{2}$	0	0
$a < \frac{1}{2}$	$1-(x/R)^{1-2a}$	1

For $a \geq \frac{1}{2}$ the inclusion of Lemma 4 (an upper bound) already suffices; for $a < \frac{1}{2}$ one needs the Claim to upgrade the inclusion to an equality.

Part II. The infimum $\inf_t X_t$

(2) $a = \frac{1}{2} \;\Rightarrow\; \inf_t X_t = 0$ a.s., and it is not attained

Idea. For $a = \frac{1}{2}$ every level $y < x$ is reached with probability 1 (Lemma 2(a)), yet 0 itself is never reached by (1).

Proof. By (1), $X_t > 0$ for all t. By Lemma 2(a) with $a = \frac{1}{2}$, $\mathbf{P}\{\tau_y < \infty\} = 1$ for each $y < x$. So a.s. $\tau_{1/n} < \infty$ for all $n > 1/x$, i.e. $\inf_t X_t \leq 1/n$ for every large n. Thus $\inf_t X_t = 0$, and the infimum is not attained. □

The other regimes.

- $a < \frac{1}{2}$: on $[0, T_x]$ the infimum is 0 and it is attained at T_x (by (4) and continuity).
- $a > \frac{1}{2}$: $\inf_t X_t > 0$ and it is attained. This is *not* a consequence of $T_x = \infty$ alone (for $a = \frac{1}{2}$ we have $T_x = \infty$ and $\inf = 0$); it follows from Part III: if $X_t \to \infty$, then $X > x$ after a random time S, and on $[0, S]$ the positive continuous function X has a positive minimum.

Part III. The limit $\lim_t X_t$

(3) $a > \frac{1}{2} \;\Rightarrow\; X_t \to \infty$ a.s.

Idea. For $a > \frac{1}{2}$, coming back down from level $2^k m$ to level m has probability $(2^{-k})^{2a-1}$, which is summable in k; by Borel–Cantelli only finitely many returns occur.

Proof. By (1), X is defined and positive for all t. Since $\{X_t \to \infty\} = \bigcap_{m\geq 1}\{\liminf_{t\to\infty} X_t \geq m\}$, it suffices to show, for each integer $m \geq 1$,

$$\mathbf{P}\Big\{ \liminf_{t\to\infty} X_t \geq m \Big\} = 1. \tag{1}$$

Fix m, let $N_k = 2^k m$, and take k so large that $N_k > x$. By Lemma 2(b), $\tau_{N_k} < \infty$ a.s. Let

$$A_k = \Big\{ \inf_{t\geq \tau_{N_k}} X_t < m \Big\}.$$

Since $X_{\tau_{N_k}} = N_k > m$ and X is continuous, on A_k the path restarted at τ_{N_k} reaches level m. By the strong Markov property at τ_{N_k} and Lemma 2(a),

$$\mathbf{P}(A_k) \leq \mathbf{P}^{N_k}\{\tau_m < \infty\} = \Big(\frac{m}{N_k}\Big)^{2a-1} = \big(2^{-(2a-1)}\big)^k.$$

Since $a > \frac{1}{2}$, $\sum_k \mathbf{P}(A_k) < \infty$. By the (first) Borel–Cantelli lemma, which needs no independence, $\mathbf{P}(A_k \text{ i.o.}) = 0$, i.e. $\mathbf{P}\big(\bigcup_n \bigcap_{k\ge n} A_k^c\big) = 1$. So for almost every ω there is a *random* $K(\omega)$ such that A_k^c holds for all $k \ge K(\omega)$. (This is not "there is a k with $\mathbf{P}(A_k^c) = 1$": K depends on ω.) For such ω, with $k = K(\omega)$ (enlarged so that $N_k > x$), $X_t \ge m$ for all $t \ge \tau_{N_k} < \infty$. This proves (1). □

$a = \frac{1}{2}$: the undecided entries of Table 1

For $a = \frac{1}{2}$ we know $T_x = \infty$ and $\inf_t X_t = 0$ (parts (1), (2)), but nothing yet about $\liminf$, $\limsup$, $\lim$. The question marked in Table 1 is whether X also goes to high levels infinitely often, or could stay below an upper barrier M while oscillating down to 0.

Theorem 5. *If $a = \frac{1}{2}$, then a.s. $\liminf_{t\to\infty} X_t = 0$ and $\limsup_{t\to\infty} X_t = \infty$. In particular $\lim_{t\to\infty} X_t$ does not exist.*

Idea. By Lemma 2, from any position the process reaches any lower level ε with probability 1 (it is recurrent) *and* any higher level R with probability 1 (no upper barrier can trap it, since it never hits 0). By the Markov property, this holds again after every deterministic time n.

Proof. By (1), $X_n > 0$ for every n. Fix $\varepsilon > 0$ and $n \in \mathbb{N}$. Condition on $\mathcal{F}_n$: if $X_n \le \varepsilon$ there is nothing to prove; if $X_n > \varepsilon$, the Markov property and Lemma 2(a) (with $a = \frac{1}{2}$) give that the restarted process reaches ε with probability 1. Hence $\mathbf{P}\{\exists s \ge n : X_s \le \varepsilon\} = 1$ for every n. Intersecting over $n \in \mathbb{N}$ and then over $\varepsilon = 1/j$, a.s. $\liminf_{t\to\infty} X_t = 0$.

Similarly fix $R > 0$: if $X_n \ge R$ there is nothing to prove; if $X_n < R$, Lemma 2(b) says that the restarted process reaches R with probability 1 (this is where $a \ge \frac{1}{2}$, i.e. that 0 is never hit, is used). So $\mathbf{P}\{\exists s \ge n : X_s \ge R\} = 1$ for all n, and intersecting over n and over $R \in \mathbb{N}$, a.s. $\limsup_{t\to\infty} X_t = \infty$. □

Remark. Another proof of $\limsup = \infty$ uses scaling: $X_t \overset{d}{=} \sqrt{t}\, X_1^1$ (Lemma 1.20), so $\mathbf{P}(X_n > R) = \mathbf{P}(X_1^1 > R/\sqrt{n}) \to 1$, and the reverse Fatou lemma gives $\mathbf{P}(X_n > R \text{ i.o.}) \ge \limsup \mathbf{P}(X_n > R) = 1$.

5 Conclusion: Table 1 completed

regime	T_x	$\inf_t X_t$	$\lim_t X_t$	$\liminf$ / $\limsup$
$a < \frac{1}{2}$	$< \infty$	0 (attained at T_x)	0 as $t \uparrow T_x$	0
$a = \frac{1}{2}$	∞	0 (not attained)	does not exist	0 / ∞
$a > \frac{1}{2}$	∞	> 0 (attained)	∞	∞

Table 2. The result: the entries ? of Table 1 are $\limsup = \infty$ and "the limit does not exist".

The conjecture of Table 1 is confirmed, and the previously undecided entries are resolved: for $a = \frac{1}{2}$, $\limsup X_t = \infty$ (there is no upper barrier), so X oscillates forever between arbitrarily small and arbitrarily large values.

Three properties of different strength. Table 2 separates three properties of a path that are easily confused:

$$X_t \to \infty \implies \inf_t X_t > 0 \implies T_x = \infty,$$

with neither arrow reversible: $a = \frac{1}{2}$ has $T_x = \infty$ but inf $= 0$; and inf > 0 alone does not give $X \to \infty$ without an argument for the upward behavior. They correspond to point transience, "not neighborhood recurrent", and transience; $a > \frac{1}{2}$ has all three, $a = \frac{1}{2}$ only the first.

Back to Brownian motion. For integer d, Table 2 is the radial form of Theorem 1: $d = 1$ ($a = 0 < \frac{1}{2}$): W hits the origin; $d = 2$ ($a = \frac{1}{2}$): W never hits the origin but returns to every neighborhood of it, infinitely often, at arbitrarily large times; $d \geq 3$ ($a > \frac{1}{2}$): $\|W_t\| \to \infty$. Moreover, Lemma 2(a) gives $\mathbf{P}\{X \text{ ever reaches level } y\} = (y/x)^{2a-1} = (y/x)^{d-2}$ for $d > 2$, the classical probability that d-dimensional Brownian motion started at distance x ever enters the ball of radius y. Proposition 1.21 extends this picture to every real $d > 0$, with $d = 2$ still the critical dimension.